

FORM 1 THE PATENTS ACT 1970 (39 of 1970) and THE PATENTS RULES, 2003 APPLICATION FOR GRANT OF PATENT (See section 7, 54 and 135 and sub-rule (1) of rule 20)				(FOR OFFICE USE ONLY)			
				Application No.			
				Filing date:			
				Amount of Fee paid:			
				CBR No:			
				Signature:			
1. APPLICANT'S REFERENCE / IDENTIFICATION NO. (AS ALLOTTED BY OFFICE)							
2. TYPE OF APPLICATION [Please tick (✓) at the appropriate category]							
Ordinary (✓)		Convention ()		PCT-NP ()		PPH ()	
Divisional()	Patent of Addition ()	Divisional()	Patent of Addition ()	Divisional()	Patent of Addition ()		
3A. APPLICANT(S)							
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3B. CATEGORY OF APPLICANT [Please tick (✓) at the appropriate category]						
Natural Person ()		Other than Natural Person (✓)			Educational institution (✓)	
		Small () Entity	Startup ()	Others ()		
4. INVENTOR(S) [Please tick (✓) at the appropriate category]						
Are all the inventor(s) same as the applicant(s) named above?		Yes ()		No (✓)		
If "No", furnish the details of the inventor(s)						
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					Country	India
					Pin code	641032

5. TITLE OF THE INVENTION

"An AI-Driven Alumni-Student Engagement Portal using Vector-Based Similarity Matching and Real-Time Asynchronous Communication"

6. AUTHORISED REGISTERED PATENT AGENT(S)

IN/PA No.	
Name	
Mobile No. (OTP verification mandatory-will be redacted)	

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8. IN CASE OF APPLICATION CLAIMING PRIORITY OF APPLICATION FILED IN CONVENTION COUNTRY, PARTICULARS OF CONVENTION APPLICATION

Country	Application Number	Filing date	Name of the applicant	Title of the invention	IPC (as classified in the convention country)

9. IN CASE OF PCT NATIONAL PHASE APPLICATION, PARTICULARS OF INTERNATIONAL APPLICATION FILED UNDER PATENT CO-OPERATION TREATY (PCT)

International application number	International filing date

10. IN CASE OF DIVISIONAL APPLICATION FILED UNDER SECTION 16, PARTICULARS OF ORIGINAL (FIRST) APPLICATION

Original (first) application No.	Date of filing of original (first) application

11. IN CASE OF PATENT OF ADDITION FILED UNDER SECTION 54, PARTICULARS OF MAIN APPLICATION OR PATENT

Main application/patent No.	Date of filing of main application

12. DECLARATIONS

(i) Declaration by the inventor(s)

(In case the applicant is an assignee: the inventor(s) may sign herein below or the applicant may upload the assignment or enclose the assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period).

We, the above named inventor(s) are the true & first inventor(s) for this Invention and declare that the applicant(s) herein are our assignee or legal representative.

(a) Date 27/03/2026

(b) Signature(s)

(c) Name(s)

Mr. ILAYARAJA N

LAVANYA M

NAVFHAL RIZWAN M

SUGANTHAN M

Yamuna
YAMUNADEVI R

(ii) Declaration by the applicant(s) in the convention country

(In case the applicant in India is different than the applicant in the convention country: the applicant in the convention country may sign herein below or applicant in India may upload the assignment from the applicant in the convention country or enclose the said assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period)

We, the applicant(s) in the convention country declare that the applicant(s) herein are our assignee or legal representative.

(a) Date 27/03/2026

(b) Signature(s)

(c) Name(s) of the signatory

(iii) Declaration by the applicant(s)

I/We the applicant(s) hereby declare(s) that: -

- ✓ We are in possession of the above-mentioned invention.
- ✓ The complete specification relating to the invention is filed with this application.
- x The invention as disclosed in the specification uses the biological material from India and the necessary permission from the competent authority shall be submitted by us before the grant of patent to me/us.
- ✓ There is no lawful ground of objection(s) to the grant of the Patent to us.
- x we are the true & first inventor(s).
- ✓ we are the assignee or legal representative of true & first inventor(s).
- x The application or each of the applications, particulars of which are given in Paragraph-8, was the first application in convention country/countries in respect of my/our invention(s).
- x I/We claim the priority from the above mentioned application(s) filed in convention country/countries and state that no application for protection in respect of the invention had been made in a convention country before that date by me/us or by any person from which I/We derive the title.
- x My/our application in India is based on international application under Patent Cooperation Treaty (PCT) as mentioned in Paragraph-9.
- x The application is divided out of my /our application particulars of which is given in Paragraph-10 and pray that this application may be treated as deemed to have been filed on DD/MM/YYYY under section 16 of the Act.
- x The said invention is an improvement in or modification of the invention particulars of which are given in Paragraph-11.

13. FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION

(a) Form 2

Item	Details	Fee	Remarks
Complete/ provisional specification)#	No. of pages 13	-	-
No. of Claim(s)	No. of claims and No. of Pages 10&2	-	-
Abstract	No. of pages 1	-	-
No. of Drawing(s)	No. of drawings and No. of Pages 6&4	-	-

In case of a complete specification, if the applicant desires to adopt the drawings filed with his provisional specification as the drawings or part of the drawings for the complete specification under rule 13(4), the number of such pages filed with the provisional specification are required to be mentioned here.

- (b) Complete specification (in conformation with the international application)/as amended before the International Preliminary Examination Authority (IPEA), as applicable (2 copies).
- (c) Sequence listing in electronic form
- (d) Drawings (in conformation with the international application)/as amended before the International Preliminary Examination Authority (IPEA), as applicable (2 copies).
- (e) Priority document(s) or a request to retrieve the priority document(s) from DAS (Digital Access Service) if the applicant had already requested the office of first filing to make the priority document(s) available to DAS.
- (f) Translation of priority document/Specification/International Search Report/International Preliminary Report on Patent ability.
- (g) Statement and Undertaking on Form 3
- (h) Declaration of Inventorship on Form 5
- (i) Power of Authority
- (j).....

Total Fee: ₹4,500/- paid via Banker's Cheque / Bank Draft bearing No. 930067 dated 27/03/2026, drawn on Karur Vysya Bank.

We hereby declare that, to the best of our knowledge, information, and belief, the facts and matters stated herein are true and correct. We respectfully request that a patent be granted to us for the said invention.

Dated this 27th day of March, 2026.

Signature:

Name:

Mr. ILAYARAJA N

LAVANYA M

NAVFHAL RIZWAN M

SUGANTHAN M

Yamuna
YAMUNADEVI R

To,

The Controller of Patents

The Patent Office, at Chennai.

Note: -

- * Repeat boxes in case of more than one entry.
- * To be signed by the applicant(s) or by authorized registered patent agent otherwise where mentioned.
- * Tick (✓)/cross (x) whichever is applicable/not applicable in declaration in paragraph-12.
- * Name of the inventor and applicant should be given in full, family name in the beginning.
- * Strike out the portion which is/are not applicable.
- * For fee: See First Schedule.

FORM 2**THE PATENT ACT 1970
(39 of 1970)****&****The Patents Rules, 2003****PROVISIONAL/COMPLETE SPECIFICATION****(See section 10 and rule13)****1. TITLE OF THE INVENTION**

Tamil-Supported AI Sign Language Interpreter for Hearing Learners

2. APPLICANT (S)

(a) NAME: Hindusthan College of Engineering and Technology

(b) NATIONALITY: Indian

(c) ADDRESS: Department of Information Technology, Hindusthan College of Engineering & Technology, Coimbatore-641032.

3. PREAMBLE TO THE DESCRIPTION**PROVISIONAL**~~The following specification describes the invention.~~**COMPLETE**

The following specification particularly describes the invention and the manner in which it is to be performed.

4. DESCRIPTION (Description shall start from next page.)**5. CLAIMS** (not applicable for provisional specification, Claims should start with the preamble — "I/we claim" on separate page)**6. DATE AND SIGNATURE** (to be given at the end of last page of specification)**7. ABSTRACT OF THE INVENTION** (to be given along with complete specification on separate page)**Note: -**

*Repeat boxes in case of more than one entry.

*To be signed by the applicant(s) or by authorized registered patent agent.

*Name of the applicant should be given in full family name in the beginning .

*Complete address of the applicant should be given stating the postal index no./code state and country.

*Strike out the column which is/are not applicable.

AI-Driven Alumni–Student Engagement Portal using Vector-Based Similarity Matching and Real-Time Asynchronous Communication

1. ABSTRACT

In the contemporary academic and professional landscape, the transition from institutional learning to industry-ready careers remains a significant challenge. While most educational institutions maintain alumni directories, these conventional systems function as passive, static databases that offer limited utility for active mentorship or career alignment. This project introduces the AI-Driven Alumni–Student Connect Portal, a sophisticated, data-driven ecosystem designed to transform institutional networking into a proactive professional engine. The platform aims to bridge the gap between academic knowledge and industry expectations by creating structured professional interactions and mentorship opportunities.

The core innovation of this research lies in its Intelligent Matching Engine, which moves beyond simple keyword filtering. By leveraging Natural Language Processing (NLP) and TF-IDF Vectorization, the system transforms raw professional profiles into high-dimensional numerical feature vectors. A Cosine Similarity algorithm is then applied to calculate the compatibility between a student's career aspirations and an alumnus's expertise, ensuring mentorship pairings are based on skill alignment and reducing the "discovery delay" found in manual networking. This automated recommendation approach improves the efficiency and accuracy of professional networking within academic institutions.

The technical architecture is built on a robust, decoupled Full-Stack framework. The React.js frontend provides a responsive, role-based user interface, while the Django REST Framework (DRF) manages business logic and secure JWT-based authentication. Data persistence is handled by a MySQL database, ensuring the integrity of professional portfolios and financial records, and WebSockets via Django Channels facilitate real-time communication between stakeholders. This architecture ensures scalability, maintainability, and secure communication across the platform.

Beyond networking, the system incorporates a Verified Institutional Fund Collection Module that addresses the lack of transparency in traditional university fundraising by requiring administrative audit workflows and document verification before any financial requisition is published. This module ensures financial accountability and builds trust among alumni contributors and institutional administrators.

Ultimately, this project provides a scalable, secure, and intelligent solution to institutional engagement by shifting from a "search-based" directory to an "AI-recommended" networking platform, improving matching accuracy, strengthening alumni-student bonds, and providing a transparent framework for institutional resource management.

The implementation of the AI-Based Analytical Engine follows a pipeline of data transformation and mathematical validation. Raw professional data—comprising skills, project descriptions, and career goals—undergoes preprocessing involving NLTK-based tokenization and noise removal. By applying TF-IDF vectorization, the system constructs a global vocabulary and maps user profiles into a high-dimensional vector space. The matching logic then employs Cosine Similarity to measure the angle between student and alumni vectors, ensuring that even if users use different phrasing for similar skills, the engine identifies the underlying professional correlation. This mathematical recommendation model significantly enhances the precision of mentorship pairings.

Furthermore, the system prioritizes Data Integrity and Security through a multi-layered authentication and audit framework. The backend implements a Stateless JWT architecture to ensure secure API verification, and every financial requisition must undergo mandatory administrative audit and KYC verification before becoming visible to the network, creating a trusted environment that encourages alumni engagement and transparent resource management within the university ecosystem. The overall system is designed to be scalable for deployment across multiple institutions and adaptable to different academic domains.

3. BACKGROUND OF INVENTION

The present invention relates generally to the fields of **educational technology (EdTech)**, **artificial intelligence**, and **professional networking ecosystems**. More particularly, the invention pertains to a mobile and cloud-enabled system and method for **automated mentorship discovery**, **career-alignment assessment**, and **institutional resource management** using AI-driven text vectorization and Natural Language Processing (NLP). The invention further relates to the real-time mathematical analysis of professional profiles and skill trajectories to generate objective compatibility metrics, including **similarity scores**, **skill-gap identification**, and **ranked mentorship matching**.

In one aspect, the invention falls within **AI-assisted career analytics** and **digital coaching platforms** that replace or augment subjective manual networking by students with standardized, data-driven matching mechanisms. The system processes unstructured textual data captured through a responsive web interface, extracts professional keypoints, derives high-dimensional feature vectors using **TF-IDF**, and applies **Cosine Similarity** logic to assess the alignment between a student's aspirations and an alumnus's expertise. The invention is applicable to both university career services and global alumni networks, especially where access to direct professional guidance is limited.

In another aspect, the invention relates to **integrated alumni development ecosystems** combining career performance analysis, longitudinal professional tracking, verified institutional crowdfunding, and secure real-time communication modules. The invention also includes mechanisms for **event-driven notifications**, **administrative audit workflows** for financial transparency, and scalable backend orchestration for near real-time feedback delivery. Accordingly, the invention lies at the intersection of **automated talent discovery**, **intelligent professional assistance**, and **secure institutional crowdfunding platforms** intended for deployment across universities, colleges, and distributed professional networks.

3. BACKGROUND OF INVENTION

Conventional alumni management and student mentorship methods depend heavily on manual networking, word-of-mouth recommendations, and static database searches, which introduce variability, subjectivity, and limited scalability across large institutional environments. In many cases, professional guidance decisions are influenced by personal connections or superficial keyword matches rather than standardized skill-based evidence, making it difficult to pair students with the most compatible mentors across diverse industries or geographical regions. Existing technology-assisted approaches also have limitations: traditional web directories function as passive repositories with no proactive engagement logic, LinkedIn-style platforms suffer from information overload and lack institutional focus, and manual career-counseling setups require significant human resources and administrative expertise that are not practical for high-volume student populations.

With the growing adoption of AI-enabled cloud computing and the shift toward digital-first professional ecosystems, there is a need for an accessible, scalable, and objective system that can assess professional alignment and mentorship potential without manual intervention. Current mainstream networking apps generally emphasize social engagement and broad tracking but lack robust, real-time, skill-specific similarity analysis and standardized scoring for talent discovery. Accordingly, a technical gap exists for an integrated invention that combines NLP-based text vectorization, automated profile similarity ranking, real-time communication protocols, and verified institutional crowdfunding in a deployable platform suitable for both academic and professional university ecosystems.

4. DESCRIPTION

I. User Authentication and Profile Management Module

This module establishes and manages the digital identity of each student, alumnus, and administrator using secure JWT-based account creation, login, and session handling. It stores structured profile attributes such as academic department, graduation year, current professional role, technical skill sets, and industry interests required for contextual matching. The module handles the

cycle, including profile updates and credential recovery, while enforcing Role-Based Access Control (RBAC) across all features. Every mentorship request, job posting, fund contribution, and matching result is mapped to this identity layer, serving as the anchor for system-wide traceability and data privacy.

II. Professional Profile and Portfolio Submission Module

This module provides the end-to-end pipeline for capturing and validating professional data from users. It enforces data entry rules for skill tags, project descriptions, and experience chronologies to ensure that the downstream AI matching remains reliable. For alumni, it includes specific workflows for posting job opportunities and mentorship availability, while for students, it focuses on career aspirations and learning goals. The module handles media preparation for document uploads (e.g., resumes or certificates), including format normalization and secure cloud storage, ensuring that the profile quality is standardized before the AI processing begins.

III. AI-Driven Analytical and Vectorization Module

This module transforms raw textual profile data into machine-interpretable mathematical signals using Natural Language Processing (NLP). It performs data cleaning (tokenization, stop-word removal, and normalization) and applies TF-IDF Vectorization to convert professional descriptions into high-dimensional feature vectors. The module computes the mathematical correlation between user profiles, identifying overlaps in technical expertise and career trajectories. By standardizing diverse phrasing into a unified vector space, this module directly improves the precision of professional discovery and reduces the noise associated with traditional keyword-based searches.

IV. Performance Matching and Similarity Evaluation Module

This module converts vectorized profile data into standardized compatibility outputs using the Cosine Similarity algorithm. It calculates a similarity score (0 to 1) for each student-alumnus pair, ranking potential mentors based on their mathematical alignment with a student's goals. The engine supports configurable thresholds—for example, only displaying matches with a score above

0.70

—to maintain a high standard of recommendation quality. In addition to numerical scores, the module identifies "aligned keywords" to provide explainable feedback, showing users exactly why a specific match was suggested.

V. Real-Time Communication and Messaging Module

This module manages instantaneous, full-duplex engagement between matched users using WebSockets and Django Channels. It facilitates private chat sessions, allowing students and alumni to interact without the delays of traditional email. The module handles message persistence, delivery status (sent/read), and real-time notifications for incoming requests. By keeping communication within a secure institutional environment, it ensures that professional networking remains focused, auditable, and protected from the external spam typical of public social media platforms.

VI. Institutional Fund Collection and Audit Module

This module introduces a transparent financial layer for institutional support and student aid. It enables the creation of verified fund requests, where each entry must include mandatory banking details (IFSC, Account Number) and supporting documentation. Unlike public crowdfunding, this module incorporates a Mandatory Administrative Audit workflow; no request becomes visible to the alumni network until it is vetted by a system administrator. This ensures financial integrity, prevents fraudulent requisitions, and builds trust within the alumni community regarding their contributions.

and Opportunity Management Module

This module enables a structured marketplace for professional opportunities within the institution. Alumni can define job metadata such as title, required skills, location, and application deadlines. The module handles the publication of these roles, their association with specific academic departments, and the tracking of student interest. By automating the distribution of opportunities to students with matching skill vectors, the module creates a localized, high-conversion talent pipeline that benefits both the hiring alumni and the seeking students.

VIII. Engagement Analytics and Trend Monitoring Module

This module applies behavioral and data analytics to track the professional "health" of the institutional network. It aggregates data on high-demand skills, mentorship frequency, and employment trends among the alumni base. It generates visual reports for administrators, identifying which industries are currently most active among their graduates. This module allows the institution to adapt its curriculum or career services based on real-world industry shifts, making the portal not just a networking tool, but a strategic asset for institutional planning.

IX. System Security and Audit Logging Module

This module ensures the platform remains compliant with data protection standards by implementing a comprehensive security framework. It handles data encryption at rest and in transit, maintains detailed audit logs of all administrative actions (especially fund approvals), and prevents unauthorized API access through stateless token validation. By separating sensitive financial data from public profile information and maintaining a clear "paper trail" of all system modifications, this module ensures that the portal is a secure environment for long-term institutional data management.

Applications

1. Automated Career Mentorship

The system can be used by universities and departments to automatically pair current students with the most compatible alumni mentors using objective skill-matching metrics instead of random networking.

2. Institutional Talent Discovery

Alumni and recruiters can evaluate student portfolios, technical skill sets, and project histories through standardized AI rankings to identify top talent for internships and job placements.

3. Remote Professional Guidance

Students can seek career advice and receive AI-assisted profile feedback from any location, enabling structured professional support even when in-person career counseling is unavailable.

4. University Placement Screening

Placement cells and career centers can deploy the platform during recruitment drives to conduct consistent and scalable candidate screening based on industry-specific skill vectors.

5. Verified Institutional Crowdfunding

The invention supports secure fundraising for campus infrastructure or student aid with automated administrative auditing, document verification, and transparent contribution tracking.

Academic boards can use aggregated alumni industry trends and student skill data to identify curriculum gaps and align educational outcomes with current market demands.

7. Longitudinal Alumni Tracking

Educational institutions can utilize the system for periodic professional tracking of their graduates, monitoring career progression, and maintaining long-term institutional engagement.

8. Real-Time Networking and Events

The platform can be used to organize digital networking events with automated matchmaking, real-time messaging, and instant notification updates for webinars or guest lectures.

9. Professional Trend Research

Researchers and administrators can utilize anonymized professional trends and employment outputs for workforce studies, comparative institutional analysis, and evidence-based career service design.

10. Inclusive and Global Alumni Access

With secure cloud hosting and multi-role support, the system can be deployed across geographically diverse alumni chapters, improving participation in technology-enabled institutional growth.

5. EXPLANATION OF THE DRAWINGS

Figure 1 – System Architecture Diagram

The architecture is designed as a layered professional networking platform in which the React-based web application communicates through a centralized Django REST API Gateway. The gateway routes requests to core domain services such as JWT authentication, role-based profile management, and mentorship matching, while enforcing strict institutional access controls. Operational data generated by these interactions is stored in a MySQL relational database to support low-latency application workflows and ensure data integrity across professional records.

For intelligence-driven networking, uploaded resumes and profile descriptions are handled by a document ingestion service and persisted in secure cloud storage. An AI processing pipeline consumes this textual data, performs Natural Language Processing (NLP) and vector-based skill analysis, and generates high-dimensional embeddings that are stored in a specialized feature store. This enables reusable, model-ready data for downstream compatibility scoring, career path comparisons, and automated student-to-mentor evaluation use cases.

In parallel, external academic records are integrated through ETL jobs and consolidated in a data warehouse for large-scale historical and cross-source professional analytics. The analytics service combines warehouse data with AI-derived profile features to compute networking insights, employment trends, and skill-gap metrics, which are delivered to an administrative dashboard for institutional decision support. Across all layers, observability components provide real-time monitoring and tracing, ensuring reliable operation of both the matching engine and the secure financial audit paths.

Figure 2 – Data Flow Diagram

The data flow diagram shows how user actions in the web client are converted into authenticated requests, NLP processing tasks, and final mentorship insights. The process starts when the end user provides professional details and career aspirations through the React interface. The frontend sends an authentication request to the backend identity service, and once validated via JWT, it

profile data, job postings, and fund-related information with the cloud database. This creates the operational context for before any AI-driven similarity analysis begins.

For profile analysis, the web client sends text-based skill descriptions to the backend, while the server stores the corresponding profile paths and submission metadata in the MySQL database. A profile-update trigger in the database initiates the AI processing stage. The Python-based analyzer then downloads the processed text from the database, runs TF-IDF vectorization and Cosine Similarity logic, and produces structured outputs such as compatibility scores, aligned skill indicators, and mentor rankings. This separation of user metadata, storage, and heavy mathematical analysis makes the system flow scalable and fault-tolerant.

After analysis completes, the generated similarity metrics are written back to the database, where they become available to the application for user-facing discovery features. The web client retrieves processed outputs as ranked recommendations, match percentages, and professional summaries, presenting them to the student through discovery, chat, and community views. In short, the diagram represents a closed-loop pipeline: input capture in the app, secure backend orchestration, AI-driven similarity evaluation, and mentor feedback delivery for continuous career development.

Figure 4.1 – Login Screen

The login screen is the entry point for registered students, alumni, and administrators to access the portal securely. It presents a clean form with two primary fields: institutional email address and password, alongside a secure "Keep me signed in" option.

The user enters their institutional credentials and taps "Sign In" to initiate the JWT authentication process. If the details are valid, the app generates a secure session token and redirects the user to the appropriate role-based home flow.

A "Join the Network" button is also provided for new students or alumni who do not yet have an account. This allows first-time users to move directly to the role-selection and registration process from the same landing screen.

Overall, the screen is intentionally minimal so users can quickly access their professional network without confusion, while still supporting secure onboarding for the institutional community.

Figure 4.2 – Home Dashboard Screen

The home dashboard is the primary workspace users see after logging in. It gives a quick overview of the professional network and acts as the central hub to start all major networking and audit activities. From this screen, users can navigate to important modules such as AI Discovery, Job Opportunities, Private Messages, Trends, Profile, and Community Posts. This makes it easy to switch between mentor matching, job hunting, and account management.

The dashboard is designed for fast access, featuring a "Quick Match" widget that shows the top recommended mentor or mentee based on recent AI analysis. Instead of searching, users can directly interact with their highest-value connections.

Overall, the home screen works as a professional control center that connects all core functions of the portal in one centralized, high-visibility layout.

Figure 4.3 – Opportunities (Job) Screen

The opportunities screen is used to discover and manage professional job postings within the institutional network. It shows available internships and roles where students can apply and compare their skills with alumni-led requirements.

A "Post an Opportunity" button is provided at the top for alumni users, allowing them to share a new opening with the student body. This helps alumni specifically recruit talent from their own alma mater. If no relevant jobs match the user's specific skill vector, the screen displays a tailored message suggesting profile updates to improve job-matching accuracy. This keeps the experience professional and encourages user data quality.

The opportunities screen supports both talent acquisition for alumni and career advancement for students, making it a central part of institutional employment interaction.

Figure 4.4 – Messages (Chat) Screen

The messages screen is where students and alumni communicate directly for career guidance and professional collaboration. It serves as the portal's private chat area for secure, institutionally-vetted conversations.

At the top, quick actions like "Search Contacts" and "Mentorship Circles" help users find their matched connections and start new professional dialogues easily. This makes it simple to collaborate without navigating through social noise.

The conversation list shows all active threads, highlighting new messages from mentors or mentees. If there are no messages yet, the screen guided the user back to the Discovery module to find a professional match.

Overall, the messages screen is designed for secure, simple, and organized professional communication using real-time WebSocket technology inside the portal.

Figure 4.5 – Profile Screen

The profile screen displays the user's professional identity and AI-derived networking stats in one place. It acts as the user's "Digital Resume" inside the institutional ecosystem. At the top, it shows basic details such as professional avatar, department, and graduation year. This helps other users quickly confirm the academic and industry background of the person they are viewing.

The "Skill Matrix" section highlights key professional indicators like top skill vectors, mentorship engagement, and networking points. It gives users a snapshot of their professional standing within the community.

The experience section contains profile fields such as work history, projects, and career aspirations. Users can update these details to refine the AI's matching accuracy and improve their compatibility with potential mentors.

The "Endorsements" area shows professional badges earned through successful mentorship and community contributions. Overall, the profile screen combines professional identity, AI progress tracking, and career motivation.

Figure 4.6 – Fund Request Creation Screen

The fund request screen is used to set up a new financial requisition for institutional support or student aid. It allows users to define all important aid details and upload necessary verification documents in one form.

Users enter the request title and a detailed description to explain the purpose of the financial need. This helps potential alumni donors understand the impact of their contribution before they commit to support.

The screen requires users to provide banking information and upload supporting documents for administrative review. This makes the fund request transparent and ensures that only verified institutional needs are broadcast.

Additional fields such as target amount and deadline are used to configure the goal and timeline of the campaign. These inputs ensure the request has a clear structure for both the requester and the auditor.

Overall, this screen makes it easy to organize verified crowdfunding and invite alumni to participate in a fair and transparent institutional support system.

Claim 1:

Claim 1:

A computer-implemented system for automated professional mentorship and institutional engagement, comprising: a User Authentication and Profile Management Module configured to manage JWT-based role identities; a Professional Profile Submission Module configured to capture and validate unstructured textual data; an AI-Based Analytical Engine configured to generate high-dimensional skill vectors from profile text; a Similarity Matching Module configured to compute compatibility scores using vector-distance metrics; a Real-Time Messaging Module configured to facilitate asynchronous communication via WebSockets; a Verified Fund Collection Module configured to manage institutional aid requests; and an Administrative Audit Module configured to enforce mandatory document verification for financial and profile transparency.

Claim 2:

The system as claimed in Claim 1, wherein the AI-Based Analytical Engine utilizes Natural Language Processing (NLP) and TF-IDF Vectorization to transform raw professional descriptions into numerical feature vectors, where feature importance is determined by the global frequency of technical keywords within the institutional database.

Claim 3:

The system as claimed in Claim 1, wherein the Similarity Matching Module calculates compatibility using a Cosine Similarity algorithm, deriving a normalized score between \$0\$ and \$1\$ based on the angular distance between a student's career-interest vector and an alumnus's expertise vector.

Claim 4:

The system as claimed in Claim 1, wherein the Verified Fund Collection Module enforces a "locked-state" for financial requisitions, where no request is visible to the alumni network until it is formally transitioned to a "verified" status by the Administrative Audit Module following the manual review of banking and KYC documents.

Claim 5:

The system as claimed in Claim 1, wherein the Real-Time Messaging Module maintains a persistent, full-duplex connection using Django Channels, allowing for instantaneous data exchange and notification delivery without the overhead of traditional polling-based HTTP requests.

Claim 6:

The system as claimed in Claim 1, wherein the Administrative Audit Module maintains a non-repudiable audit log of every profile modification and fund approval, ensuring traceability and institutional accountability for all high-privilege actions.

Claim 7:

The system as claimed in Claim 1, further comprising an Analytics and Trend Monitoring Module configured to aggregate anonymized profile data to generate longitudinal reports on industry demands, top-matched skills, and departmental employment success rates.

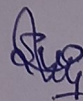
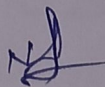
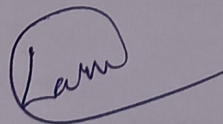
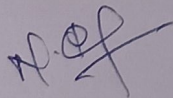
Claim 8:

The system as claimed in Claim 1, further comprising a Job and Opportunity Management Module configured to automatically broadcast alumni-posted career openings to student users whose profile vectors exceed a predefined similarity threshold relative to the job's required skill vector.

computer-implemented method for institutional mentorship matching, comprising: acquiring raw professional text; extracting technical and professional landmarks with assigned weights; computing high-dimensional skill vectors; calculating mathematical compatibility using Cosine Similarity; ranking alumni candidates; validating institutional fund requests through administrative auditing; and initiating real-time communication between matched stakeholders.

Claim 10:

A non-transitory computer-readable medium storing instructions that, when executed, cause performance of the method of Claim 9, wherein a Stateless JWT architecture ensures that security and scoring consistency are maintained across the React frontend and Django backend execution paths.



Yamuna

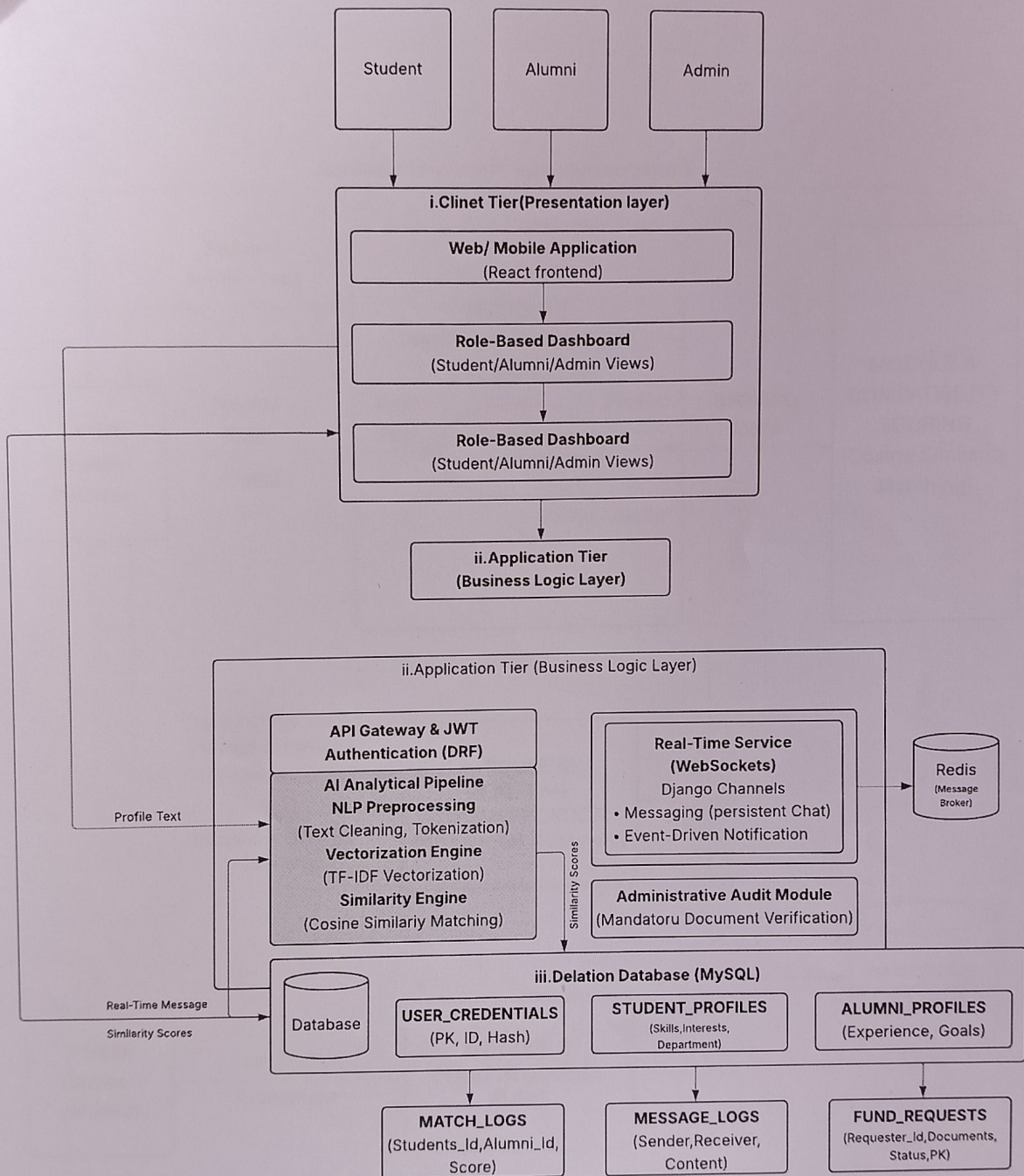


Figure 1 – System Architecture Diagram

Handwritten signatures and names at the bottom of the page:

- Top left: *Handwritten signature*
- Top middle: *Handwritten signature*
- Top right: *Handwritten signature*
- Bottom left: *Yamuna*

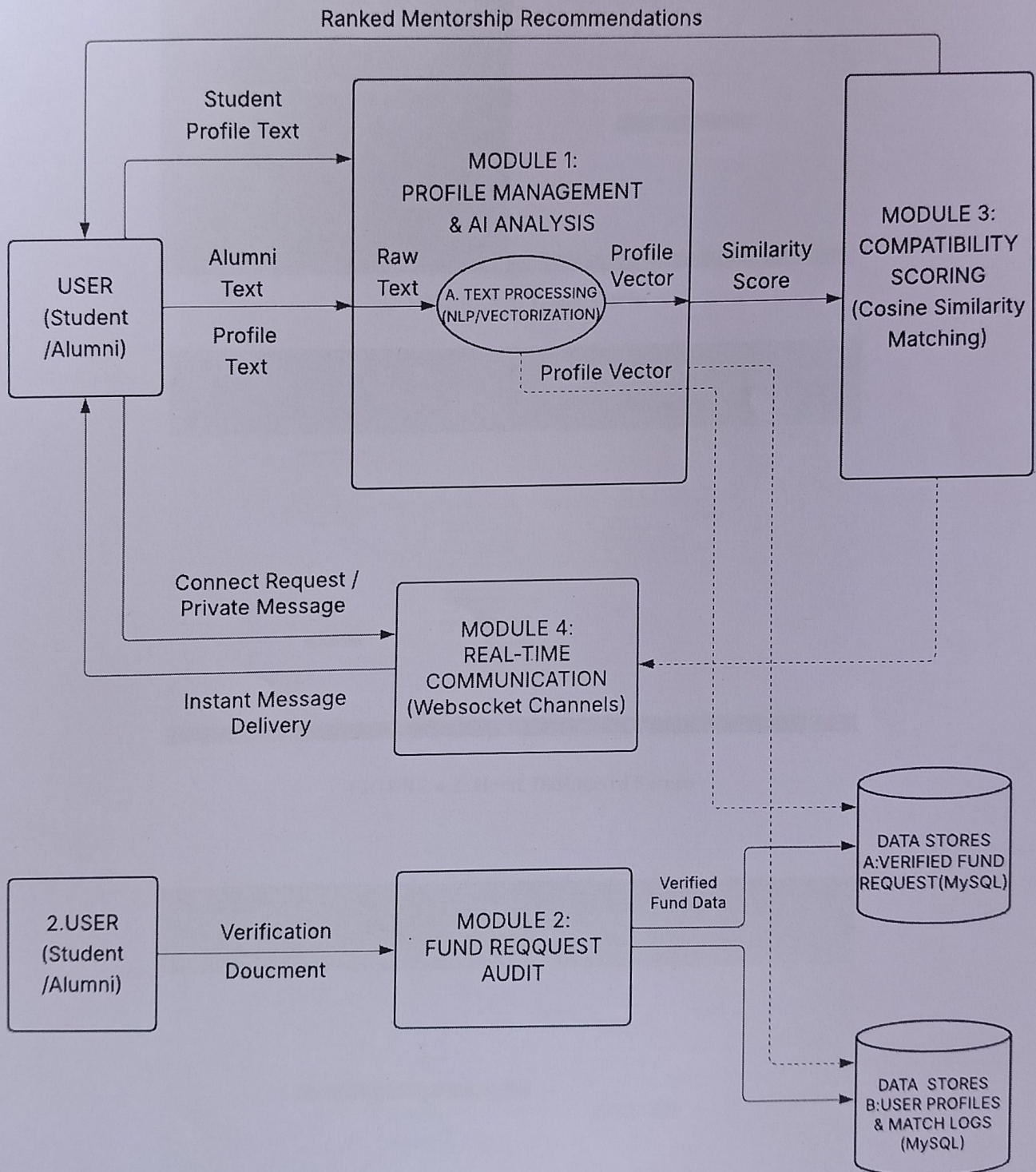


Figure 2 – Data Flow Diagram

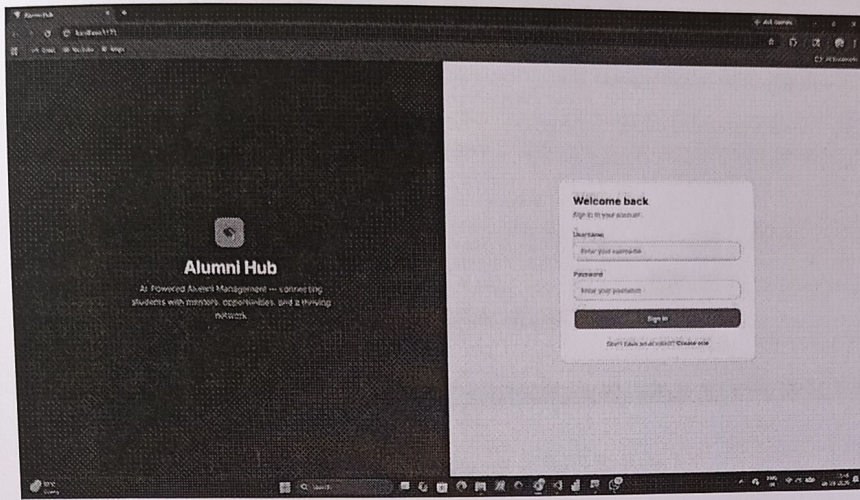


FIGURE 4.1: Login Screen

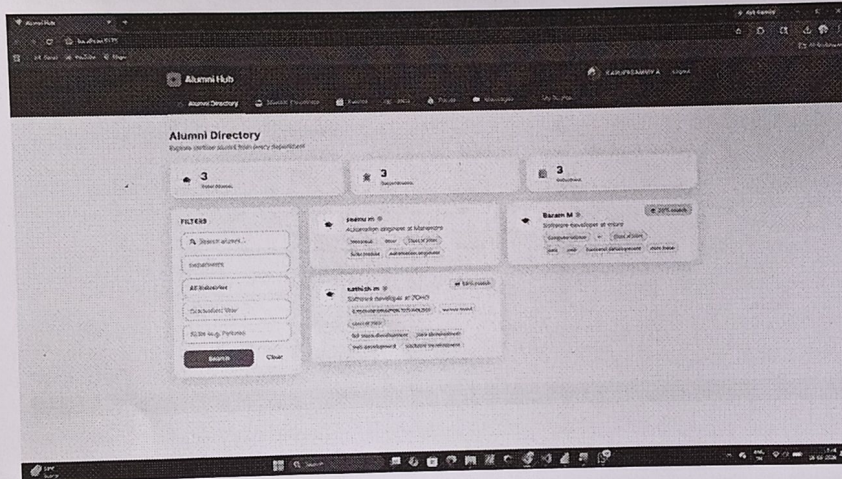


FIGURE 4.2: Home Dashboard Screen

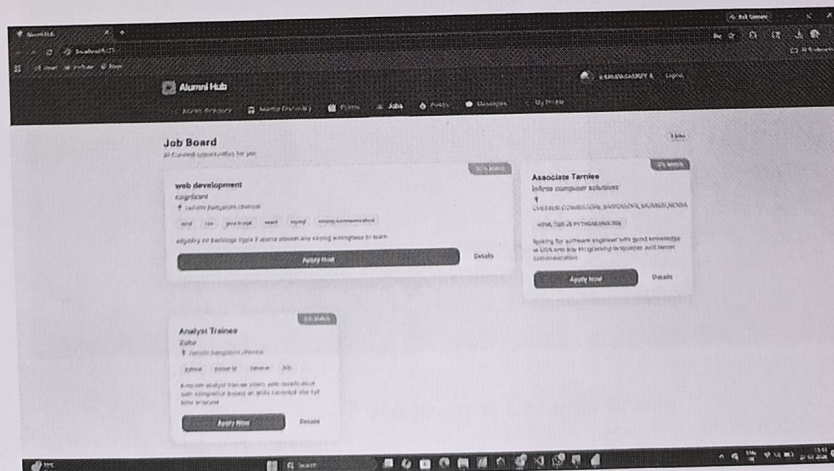


Figure 4.3 – Opportunities (Job) Screen

N-P
Kamuna
Lana
N-A
Rudra

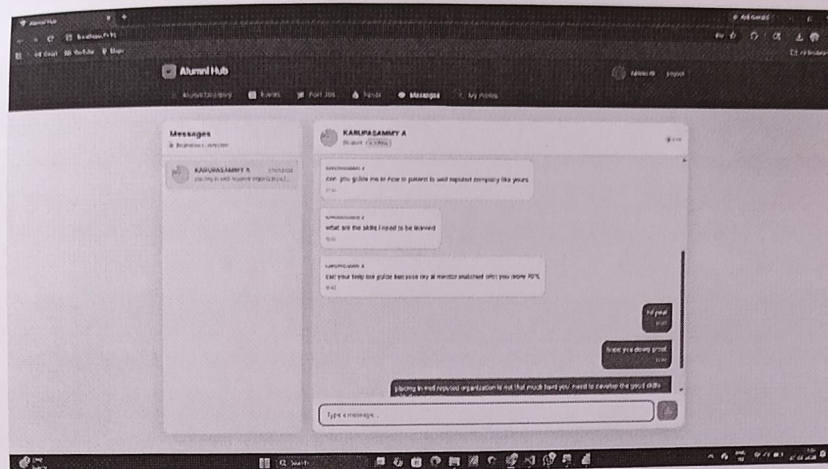


Figure 4.4 – Messages (Chat) Screen

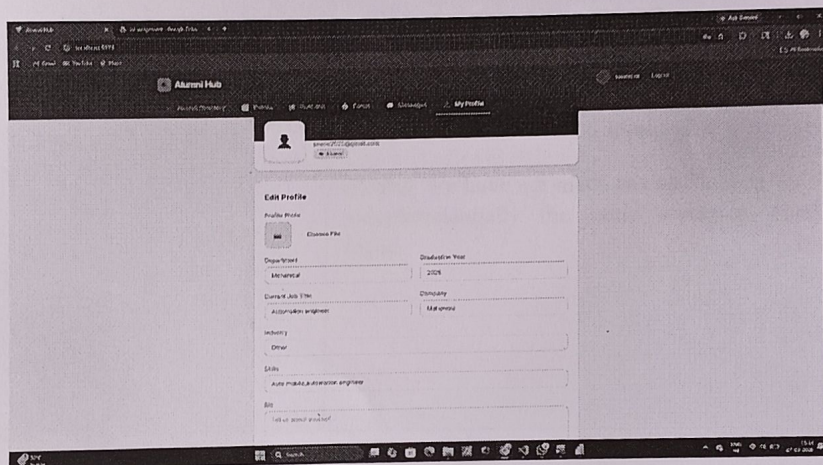


Figure 4.5 – Profile Screen

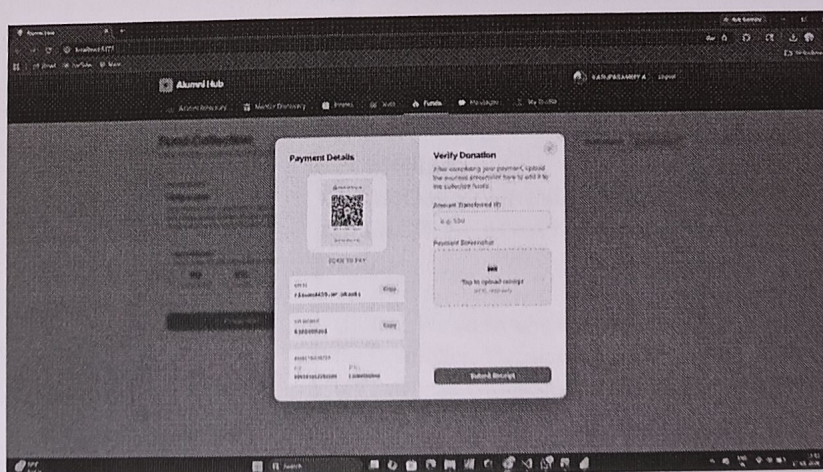


Figure 4.6 – Fund Request Creation Screen

N. C. P.
Yamuna

Lana

N. C. P.

S. C. P.

FORM 3

THE PATENTS ACT, 1970

(39 of 1970)

and

THE PATENTS RULES, 2003

STATEMENT AND UNDERTAKING UNDER SECTION 8

(See sub-rule (2) and (3) of Rule 12)

1. Name of the applicant(s).

I/We: HINDUSTHAN COLLEGE OF ENGINEERING AND
TECHNOLOGY.....
.....

hereby declare:

2. Name, address and nationality of the joint applicant.

(i) that We who have made the application for patent number
in India, dated 27/03/2026, alone,(ii) that We have not made any application for the
same/substantially the same invention outside
India.Name of
the
countryDate of
applicationApplication
No.Status of the
applicationDate of
publication

Date of disposal

3. Name and address of the assignee

that the rights in the application(s) filed in India has/have been assigned
to HINDUSTHAN COLLEGE OF ENGINEERING AND
TECHNOLOGY,
DEPARTMENT OF INFORMATION TECHNOLOGY,
POLLACHI HIGHWAY, COIMBATORE – 641032, TAMIL NADU,
INDIA.(i) that We undertake that upto the date of grant of the patent by
the Controller, We would keep him informed in writing regarding
the details of corresponding applications for patents filed outside
India in accordance with the provisions contained in section 8 and
rule 12.

Dated this 27th day of March, 2026

4. To be signed by the applicant
or his authorized registered
patent agent.

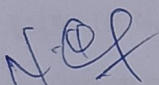
Signature.....

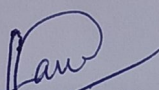
PRINCIPAL

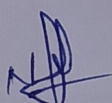
Hindusthan College Of Engineering & Technology

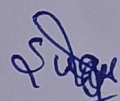
COIMBATORE - 641 032

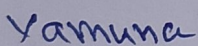
5. Name of the natural person who has signed.	MR. ILAYARAJA N LAVANYA M NAVFHAL RIZWAN M SUGANTHAN M YAMUNADEVI R
	To The Controller of Patents, The Patent Office, at.....Chennai.....
Note. - Strike out whichever is not applicable.	


Mr. ILAYARAJA N


LAVANYA M


NAVFHAL RIZWAN M


SUGANTHAN M


YAMUNADEVI R

FORM 5

THE PATENTS ACT, 1970

(39 of 1970)

&

The Patents Rules, 2003

DECLARATION AS TO INVENTORSHIP

[See section 10(6) and rule 13(6)]

1. NAME OF APPLICANT (S)

HINDUSTHAN COLLEGE OF ENGINEERING AND TECHNOLOGY

hereby declare that the true and first inventor(s) of the invention disclosed in the complete specification filed in pursuance of our application numbered dated are .

2. INVENTOR (S)

NAME MR. ILAYARAJA N

LAVANYA M

NAVFHAL RIZWAN M

SUGANTHAN M

YAMUNADEVI R

(a) NATIONALITY INDIAN

(b) ADDRESS DEPARTMENT OF INFORMATION TECHNOLOGY, HINDUSTHAN COLLEGE OF ENGINEERING AND TECHNOLOGY, COIMBATORE – 641032, TAMIL NADU, INDIA

Dated this 27th day of March, 2026

Signature: -

Name of the signatory: -MR.ILAYARAJA N

LAVANYA M

NAVFHAL RIZWAN M

SUGANTHAN M

YAMUNADEVI R

3. DECLARATION TO BE GIVEN WHEN TÄEFFLICATION IN INDIA IS FILED BY THE APPLICANT (S) IN THE CONVENTION COUNTRY:

We the applicant(s) in the convention country hereby declare that our right to apply for a patent in India is by way of assignment from the true and first inventor(s).

Dated this27...day of ...03....2026

Signature: -

Name of the signatory:

4. STATEMENT (to be signed by the additional inventor(s) not mentioned in the application form)
We assent to the invention referred to in the above declaration, being included in the complete specification filed in pursuance of the stated application.

Dated this 27 day of 03...2026.....

Signature of the additional inventor(s): Name: -

To, The Controller of Patent
The Patent Offices, at. Chennai

Note

***Repeat boxes in case of more than one entry.**

***To be signed by the applicant(s) or by authorized registered patent agent otherwise where mentioned.**

***Name of the inventor and applicant should be given in full, family name in the beginning .**

***Complete address of the inventor should be given stating the postal index no./code, state and country.**

***Strike out the column which is/ are not applicable**

FORM 9
THE PATENTS ACT, 1970
(39 of 1970)
&
The Patents Rules, 2003
REQUEST FOR PUBLICATION
[See section 11A (2); rule 24A]

1. Name, address and nationality of the applicant(s).

HINDUSTHAN COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY,
HINDUSTHAN COLLEGE OF ENGINEERING AND TECHNOLOGY, COIMBATORE – 641032, TAMIL NADU,
INDIA
INDIAN

2. To be signed by the applicant or his authorized registered patent agent.

We hereby request for early publication of our application for patent **dated 27th day of March, 2026** under section 11A(2) of the Act.

Dated this 27 day of 03, 2026

3. Name of the natural person who has signed.

Mr. ILA YARAJA N, LAVANYA M, NAVFHAL RIZWAN M,
SUGANTHAN M, YAMUNADEVI R

Signature

The Controller of Patents,
The Patent Office,
At
Chennai

Karur Vysya Bank
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Valid for three months from the date of issue

DEMAND DRAFT

MALUMICHAMPATTI, 340/1-B, INDIRA NAGAR, POLLACHI MAIN ROAD,
MALUMICHAMPATTI, COIMBATORE - 641 050
IFSC Code : KVBL0001977

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Purchaser Name: NAVTHAL RIZWAN M

AUTHORISED SIGNATORY

AUTHORISED SIGNATORY

अदाकर्ता शाखा / DRAWEE BRANCH

Please sign above

007301

CHENNAI-CENTRAL CLEARING OFFICE

430067 6410530421 005615 16

HINDUSTHAN COLLEGE OF ENGINEERING AND
TECHNOLOGY, COIMBATORE - 641032, TAMIL NADU,
INDIA
INDIAN

2. To be signed by the applicant or
his authorized registered patent
agent.

We hereby request for early publication of our
application for patent dated 27th day of March,
2026 under application no. 2026/2026